MIL-H-5035B(ASG)
26 MARCH 1958
Superseding
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29 June 1953

MILITARY SPECIFICATION

HOIST; BOMB, MANUAL, AERO 14B

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

1. SCOPE

1.1 This specification covers a manual bomb hoist, having a capacity load of 2,240 pounds, designated as Hoist, Bomb, Manual, Aero 14B and hereinafter referred to as "hoist", for use on military aircraft. This hoist will not be interchangeable with the Mk 8 Mod 0 Portable Bomb Hoist and cannot be used on bomb rack hoisting brackets.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawing, and publication, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-D-5028	Drawings and Data Lists: Preparation of Manufacturers' (for Production Aeronautical and Associated Equipment)
MIL-C-11796	Corrosion Preventive, Petrolatum, Hot Application
MIL-C-16173	Corrosion Preventive, Solvent Cutback, Cold-Application

STANDARDS

Federal

FED. TEST METHOD STD. NO. 151	Metals; Test Methods

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

No. 143	Specifications and Standards; Use of

FSC 1730

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DRAWINGS

Navy - Bureau of Aeronautics

52A226D17 List of Drawings Portable Bomb Hoist Aero 14B (and all drawings listed thereon)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample.- Prior to beginning quantity production, preproduction samples shall be subjected to preproduction testing.

3.2 Materials.- Materials used in the manufacture of the manual bomb hoist shall conform to the applicable specifications specified herein or as listed on applicable drawings, if such materials are available. Materials conforming to contractor's specifications may be used provided the specifications are released by the Government, are equivalent to the Government specifications, and contain requirements for adequate tests of the material. The use of contractor's specifications shall not constitute a waiver of Government inspection. All materials shall be of the lightest practicable weight and entirely suitable for the intended purpose. When materials are used which are procured under commercially recognized non-Government specifications, a certificate, acceptable to the Government Inspector, showing compliance with such specifications shall be furnished by the supplier of such material.

3.3 Selection of materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.3.1 Standard parts.- Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.4 Conformance to drawings.- No departure from or revision or change to the drawings specified herein shall be permitted without prior written approval by the procuring activity. In the event changes are approved, the contractor shall furnish the procuring activity with reproducible drawings covering the change. Conformance to the drawings will not relieve the contractor of the requirements of 3.5 below.

3.5 Manufacturing and finish.- Manufacturing processes and finishes shall be as shown on the drawings furnished by the procuring activity.

3.6 Interchangeability.- All parts having the same part number shall be completely interchangeable with each other with respect to installation and performance. Changes in Manufacturer's part numbers shall be governed by the drawing requirements of Specification MIL-D-5028. Parts joined together as by welding or riveting shall be interchangeable as assemblies. Proper gaging methods and facilities shall be maintained to insure that this requirement is fulfilled.

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3.7 Performance.- The hoist shall be capable of raising and lowering a capacity load of 2,240 pounds efficiently and safely. The hoist shall operate quietly and be capable of withstanding the tests prescribed in section 4.

3.8 General operating requirements.- The Aero 14B manual bomb hoist shall function properly under all tests specified in section 4.

3.9 Operating conditions.- The hoist shall be designed to function properly under any or all the following conditions.

(a) Altitude: From sea level to 6,000 feet elevation.
(b) Temperature: The hoist shall be capable of operating with no load and with all loads up to a capacity load of 2,240 pounds at temperatures of 0°F and +160°F.
(c) Humidity: Under all conditions of relative humidity encountered at the above temperatures and altitudes.
(d) Vibration: See 4.5.9.
(e) Attitude: See 4.5.3.

3.10 Identification of product.- Each hoist shall be marked for identification in accordance with Standard MIL-STD-130.

3.11 Workmanship.- The workmanship shall be that required by the best commercial practice, covering this type of equipment, for the quantity production of analogous parts. All operating parts of the hoist shall be free from imperfection which may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests.- The inspection and testing of the hoist components shall be classified as follows:

(a) Preproduction Tests (See 4.3.)
(b) Acceptance Tests (See 4.4.)

4.2 Test conditions

4.2.1 Tests conducted by the contractor shall be accomplished under prevailing conditions, unless otherwise specified. All tests conducted by the procuring activity shall be conducted under simulated service conditions. All parts of the test articles shall be stabilized at room temperature, when other than that of the laboratory, prior to the conductance of the test. Installation for test shall simulate service conditions, unless otherwise specified.

4.2.2 Auxiliary equipment used in connection with these tests shall be of standard or approved type used in service installation of the test articles and whose performance characteristics are known.

4.2.3 Pretest check.- All movable or adjustable working parts of the hoist shall be operated sufficiently prior to testing to insure that they are in proper adjustment and working condition. The hoist shall be operated first without load and then with a load of approximately 50 percent of the capacity load, (2,240 pounds), through a sufficient number of cycles of all applicable operations to insure proper breaking in of all parts. The following items shall be carefully checked over prior to testing:

Extension tube
Mechanical brake
Cable limiting device
Cable Antifouling device
Overload safety device

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4.2.4 Break-in.- During the break-in period, care shall be taken to avoid operation which will impose any severe strain on the equipment.

4.3 Preproduction Tests.-

4.3.1 Sampling instructions.- The preproduction samples shall consist of the first two hoists produced under contract and manufactured by quantity production methods and tools representative of the hoists, and shall be tested in a place and manner designated in the contract, purchase order, or invitation for bids.

4.3.1.1 Certification from the cognizant Government Inspector that the materials used in the samples are in accordance with requirements specified herein (see 3.2) shall be forwarded with the samples.

4.3.2 Test schedule.- The preproduction samples shall be tested in accordance with 4.2 and any or all other tests described in section 4 as determined by the testing activity to be necessary for determining the quality of the samples. An appropriate test schedule shall be arranged whereby the two preproduction samples will be undergoing tests simultaneously. In addition to the examination required in 4.2, each sample shall be subjected to at least one half of the other tests described in 4.5. Each test shall be conducted on at least one preproduction sample.

4.3.3 Tests.- The Preproduction Tests shall consist of all the tests of this specification.

4.4 Acceptance Tests.- The Acceptance Tests shall consist of Individual unit tests and Sampling tests.

4.4.1 Individual unit tests.- Each hoist submitted for acceptance under the contract shall be inspected and tested by the contractor in accordance with 4.5.1 and shall be operated manually or on a power-driven fixture (furnished by the contractor as approved by the Government Inspector). The hoist shall be operated through 3 consecutive cycles with capacity load of 2,240 pounds in both directions through a distance of 8 feet.

4.4.2 Sampling tests.-

4.4.2.1 Sampling instructions.- The number of units to be selected at random from each production lot for production lot samples shall be as specified below.

Quantity of units in production lot	Quantity of samples to be selected
99 or less	1
100 to 299	3
300 to 500	6

4.4.2.2 Limitation of lot size.- For sampling purposes, lots of over 500 units shall be divided into smaller lots in accordance with the above list.

4.4.2.3 Sampling test schedule.- Each production lot sample shall be subjected to the tests prescribed for individual units, (see 4.4.1). The samples shall also be subjected to the additional tests specified in table I.

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TABLE I

Sampling test schedule

Tests	Paragraph	Quantity of samples (per lot size) to be subjected to test		
		99 or less	100 to 299	300 to 500
Braking	4.5.4	1	1	1
Manual Operation	4.5.2	1	1	1
Life	4.5.5	1	1	1
Sand and dust	4.5.8	1	1	1

4.5 Tests.-

4.5.1 Examination of product.- This test shall consist of (a) a complete dimensional check to insure compliance with applicable drawings with respect to sizes of parts; (this may be done prior to or after assembly as necessary); (b) a complete check of surface finishes on all parts to insure compliance with requirements and adequacy of finishes; (c) a thorough assembly check to insure proper assembly; (d) a check of the workmanship to insure required quality. In addition to the above check items under this test, the preproduction samples and production lot samples shall be subjected to a complete material and process check to insure use of certified materials (see 3.2) and compliance with required and approved manufacturing processes, (see 3.5). Also, the preproduction samples shall be weighed to determine compliance with the specified requirements. The checks under this test may be conducted prior to or after all other required tests at the discretion of the Inspector or testing activity.

4.5.2 Manual operation.- The hoist shall be operated manually for 10 consecutive cycles by one man with a capacity load of 2,240 pounds in both directions through a distance of 8 feet. The effort to raise or lower the capacity load shall not exceed 35 pounds (17-1/2 pounds per handcrank) on a radius of 8 inches.

4.5.3 Attitude of hoist.- With the hoist attached to the hoisting bracket or stowed by hanging on a bulkhead bracket or lying horizontally in any position for a period of 1 hour on a workbench, there shall be no leakage of oil or grease as the result of this test. The normal working position of the hoist will be approximately 45 degrees to the horizontal plane.

4.5.4 Braking.- While operating the hoist under (a) 100-pound load or (b) 2,240-pound capacity load, the mechanical brake shall automatically stop and hold the load upon release of the handcranks, whether the release is intentional or accidental. There shall be no interference owing to the mechanical brake, such as binding, slipping, jerky action, etc.

4.5.5 Life.- The ability of the hoist to operate without failure or readjustment of any parts through a total of 500 cycles shall be determined. Each cycle shall consist of raising and lowering a weight 8 feet in the following sequence: Raise and lower 2,240 pounds 20 times with a rest period of 20 seconds between each cycle. Raise and lower 1,000 pounds 30 times with a rest period of 20 seconds between each cycle. These tests to be conducted in the order stated and repeated 10 times to complete the required 500 cycles.

NOTE: This test shall be conducted at the contractor's plant, using a power-driven unit in lieu of manual operation. This test need not be conducted during one continuous period of time but may be during normal working hours.

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4.5.6 Salt spray.- The hoist shall be exposed to a salt spray in accordance with Federal Test Method Standard No. 151, Method 811, for 35 hours. At the completion of the exposure, the hoist shall be stored, for at least 24 hours in a room at ambient temperature and atmospheric pressure after which it shall operate satisfactorily for 5 cycles raising and lowering a capacity load of 2,240 pounds through a distance of 8 feet. Prior to this test all exposed moving parts shall be properly lubricated as necessary.

4.5.7 Overload.- A static test load of 5,600 pounds (capacity load of 2,240 pounds x 2-1/2) shall be suspended by the hoist for 3 minutes in the same manner as that in which the capacity load of 2,240 pounds is suspended during normal operation. The cable shall be in the reeled and unreeled position. At least two full turns of cable shall remain on the drum. No part of the hoist shall take a permanent set. After the completion of the test, the hoist shall operate satisfactorily for 5 cycles, raising and lowering a capacity load of 2,240 pounds through a distance of 8 feet. This test shall be conducted at the contractor's plant.

NOTE: The used cable shall be replaced prior to Acceptance test and the cables used for overload test destroyed.

4.5.8 Sand and dust.- The completely assembled hoist shall be subjected to conditions and tests specified in paragraph 4.5.8.2. If facilities are not available for testing the hoist in the dust chamber, it is permissible to transfer the hoist to a test fixture, located at room temperature and within 3 minutes after transfer, test the ability of the hoist to meet specification requirements.

NOTE: Prior to and subsequent to these tests, all parts of the hoist shall be properly lubricated giving particular attention to those parts that might permit sand and dust penetration.

4.5.8.1 Apparatus.- The test apparatus shall consist of a test chamber capable of maintaining internal temperatures of -70°F, and +160°F and a relative humidity of not more than 70 percent. The chamber shall also be capable of maintaining air velocities of 2,300 ±50 feet per minute and 100 ±10 feet per minute and a sand and dust density of 0.30 ±0.03 grains per cubic foot within the test space. The sand and dust particles shall pass through a 170-mesh screen and not more than 50 percent by weight shall pass through a 200-mesh screen.

4.5.8.2 Procedure.- The hoist shall be placed in the test chamber and the sand and dust density raised and maintained at the value specified above. The internal temperature of the test chamber shall be maintained at -40°F for a period of 1 hour with an air velocity through the test area of 2,300 ±50 feet per minute and 1 hour with an air velocity through the test area of 100 ±10 feet per minute. The temperature shall then be raised to and maintained at +120°F for a period of 1 hour with an air velocity through the test area of 2,300 ±50 feet per minute and 1 hour with an air velocity through the test area of 100 ±10 feet per minute. At the end of the 4-hour test period, the equipment shall be allowed to cool and shall be operated and examined in accordance with 4.5.8.4 below. After completion of the test, shake off excess sand and dust by brushing and tapping vigorously. All units of the hoist shall function properly during normal operation in the following sequence.

4.5.8.3 Operation.- The hoist shall be operated to raise and lower first a load of 100 pounds and then a capacity load of 2,240 pounds in turn five times through a distance of 5 feet.

4.5.8.4 Dust test inspection.- All the vulnerable parts of the hoist shall be disassembled and in the event of any sand penetration, the necessary corrective measures taken and then the test repeated. Any penetration of dust or sand shall be considered cause for rejection of the hoist.

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4.5.9 Vibration.- The completely assembled hoist shall be vibrated in a vertical and horizontal plane for a total of 1 hour in each plane. Vibration shall follow approximately a simple harmonic motion; the frequency shall be maintained at 3,000 cycles per minute. The amplitude shall be between 0.003 and 0.005 inch (total displacement of 0.006 to 0.010 inch). Any mechanical failure or loosening of any part during the test shall be considered failure of the unit to pass the test.

4.6 Rejections and retest.-

4.6.1 Rejection of preproduction samples.- The failure of either of the preproduction samples to pass the required tests, shall be cause for tentative rejection of both the preproduction samples. At the discretion of the procuring activity, deliveries of production units will be conditionally accepted until the preproduction samples satisfactorily complete the required tests and all corrections made on the preproduction samples are incorporated in the production units.

4.6.1.1 Correction of defects in preproduction samples.- Any defects discovered during inspection and testing of the preproduction samples shall be corrected by the contractor at his expense. Such corrective measures as may be found necessary shall likewise be incorporated in the production of all subsequent units submitted for acceptance under the contract.

4.6.1.2 Retest of preproduction samples.- Preproduction samples which have been rejected and resubmitted for test shall be tested in accordance with one or more of the test methods described under 4.5 as determined by the testing activity.

4.6.2 Rejection of individual unit.- The failure of any hoist, any piece, or subassembly to pass the inspection and tests required in 4.5 shall cause its tentative rejection, subject to retest as hereinafter provided.

4.6.2.1 Retest of individual units.- Any hoist or any piece, subassembly, or assembly which fails to pass the individual unit tests as specified in 4.4.1 may be resubmitted upon the request of the contractor for individual inspection and test, provided a statement showing the corrective measures taken is furnished to the Inspector at the time of such resubmission. Upon reinspection and retest, failure to meet the requirements outlined in 4.4.1 shall be cause for final rejection of the unit, unless a second retest is specifically approved by the Government Inspector.

4.6.3 Rejection of production lot samples.- The failure of any production lot sample to pass the inspections and tests required in 4.4.2.3 shall cause tentative rejection of all units represented by that sample, subject to retest as hereinafter provided.

4.6.3.1 First retest, production lot samples.- In the event any sample representing a production lot fails to pass the inspection and test required in 4.4.2.3 herein, the contractor shall have the opportunity of reworking the entire affected lot and resubmitting to retest twice as many samples, selected at random by the Inspector, as originally submitted for the lot. Should all of these samples pass the required tests, the lot represented thereby shall be considered acceptable. Should any of these samples fail, the entire lot represented thereby shall be rejected.

4.6.3.2 Second retest, production lot samples.- If, in the contractor's opinion, the failure of the retest samples is due to a remediable defect, the contractor may, on the approval of the Inspector, be offered the opportunity of reworking the entire affected lot and resubmitting for a second retest three times as many samples, selected at random, as originally submitted for the lot. Should all of these samples pass the required tests, the lot represented thereby shall be considered acceptable. Should any of these samples fail, the entire lot represented thereby shall be finally rejected, unless specific approval for further rework and retest is granted by the procuring activity.

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4.6.4 Inspection and testing subsequent to delivery.- Any or all hoists subsequent to the first two "preproduction samples" may be subjected to inspection and testing to determine their conformance with any or all requirements and tests of this specification after delivery to the Government. Any hoist found to be defective owing to noncompliance with the requirements of this specification and applicable specifications and drawings may be rejected after delivery.

4.7 Salvaged pieces.- Any piece, or pieces, from a hoist may be salvaged for use, provided each salvaged piece is inspected for and is in complete compliance with the pertinent drawings and requirements of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packing.-

5.1.1 Preservation.- All working and bearing surfaces shall be suitably protected against corrosion during shipment and storage by suitable grease or preservatives or other approved method in accordance with Specifications MIL-P-116 and MIL-C-11796 or MIL-C-16173. The hoist shall be prepared for shipment in a manner which will facilitate quick set-up for use by the recipient with a minimum amount of necessary greasing, oiling, or adjusting of the various parts of the equipment.

5.1.2 Packing for domestic shipment.- Unless otherwise specified, the manual bomb hoist shall be packed in substantial exterior shipping containers so constructed as to insure acceptance by common or other carrier for safe transportation at lowest rates to the point of delivery specified in the contract or order. Containers shall conform to the requirements of the Consolidated Freight Classification Rules in effect at the time of shipment. Containers shall be capable of withstanding storage, rehandling, and reshipment without the necessity of repacking.

5.1.2.1 Packing of accessory parts.- Accessory parts may be detached and packed to insure against injury in transit but, if detached for separate packing, the accessories shall be secured to the hoist in a manner which will assure against their accidental separation from their respective units during transit. Detached accessories shall be preserved and packed in accordance with 5.1.1 and 5.1.2.

5.2 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The identification shall include the following information:

HOIST; BOMB, MANUAL, AERO 14B
Specification MIL-H-5035B

6. NOTES

6.1 Intended use.- The hoist is for use in hoisting military stores for attachment to bomb racks and shackles aboard various types of military aircraft.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Where the Preproduction Test samples should be sent, the activity responsible for testing, and instructions concerning the submittal of test reports. (See 3.1 and 4.2.1.)

6.3 Availability of dummy stores for test.- The contractor will submit requests to the procuring activity for dummy and or inert stores to be used for testing as required by this specification.

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6.4 Information required with bids.- Each bid will be accompanied by a statement as to whether or not any exceptions are being taken to the requirements shown in the drawings and specifications. If exceptions are being taken or deviations are anticipated, alternative proposals will be indicated in the statement.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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